Abstract

This thesis explores the design of online caching algorithms augmented with machine learning (ML) predictions. Recent years have seen growing interest in such learning-augmented algorithms which aim to outperform classical bounds by leveraging ML predictions while maintaining robust to poor predictions. For the online paging problem, existing approaches typically achieve these guarantees by dynamically switching between classical eviction rules and prediction-based decisions. We identify fundamental limitations in this paradigm - although it ensures robustness, it fails to realize the full potential of integrating classical algorithms with learned predictions.

This thesis proposes a simpler and more practical approach that combines predictions with classical rules more seamlessly. Our key insight is that these signals provide complementary information that should be combined rather than alternated between. We provide theoretical analyses showing that this weighted integration can reduce variance in the predicted distance estimates, leading to more effective eviction decisions while maintaining strong worst-case bounds similar to those in prior literature. Initial experiments on standard caching benchmarks validate this insight, demonstrating that appropriate weighted combinations significantly outperform both pure strategies across diverse workloads.

A key challenge in deploying such weighted integration is determining appropriate weights for combining the signals in an online setting. The optimal weight varies across workloads and may change dynamically as access patterns evolve. We develop and analyze two approaches: an efficient golden-section search leveraging empirically observed unimodality, and a more robust Gaussian process-based method that handles noisy observations and changing distributions. Further experimental evaluation shows our online parameter tuning methods often achieve the performance of offline optimal weight while maintaining computational efficiency.

This work suggests broader implications for algorithm design, showing how thoughtfully integrating ML predictions with classical algorithmic principles can achieve better performance than treating them as competing alternatives. We believe similar principles could improve learning-augmented algorithms for other fundamental problems beyond caching.